

ENGINEERING SPECIFICATIONS

Document Name: IO list for KWK Siemens Single Circuit		
Document No: ESP-23-24-006	Revision: R01	Date: 03/05/2023
Prepared By: PJP		Approved By: CGH

Analog Input/Output	POL688	X1	Suction Pressure Comp
		X2	Discharge Pressure Comp (4-20 mA)
		X3	Oil Feed Pressure Comp (4-20 mA)
		X4	Evaporator Water In Temperature
		X5	Evaporator Water Out Temperature
		X6	Comp Current (0-5 Vdc)
		X7	Suction Temperature
		X8	Discharge Temperature Comp
		X9	Liquid Line Temperature
		X10	Condenser Water In Temperature
		X11	Condenser Water Out Temperature
	POL985	X1	VFD command
		X2	Economiser EXV Comp (Analog Output 0-10 Vdc)
		X3	Desuperheater Out Temperature
		X4	EXV Command (Analog Output 0-10 Vdc)
		X5	Oil Discharge Pressure Comp
		X6	Evaporator Flow Meter
		X7	Condenser Flow Meter
		X8	Economiser Out Pressure Comp
		B1	Economiser Out Temperature Comp
		B2	Sump Oil Temperature
		B3	Desuperheater In Temperature

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Digital Input	POL688	D1	Evaporator Flow switch
		D2	Condenser Flow Switch
		DU1	Comp ON/OFF Switch (24V AC/DC)
		DU2	Voltage Protection Device/VFD Fault (24V AC/DC)
		DL1	Transition Feedback Comp (AC 115...230 V)
		DL2	Winding Protection Comp(AC 115...230 V)
	POL985	D1	Oil Level Switch
		D2	External Emergency Stop
		D3	SPARE
		DL1	LP Cutout
		DL2	HP Cutout

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Digital Output	POL688	Q1	Comp Alarm
		Q2	Compressor (Delta) Comp/VFD Latch Comp
		Q3	Compressor (Main) Comp
		Q4	Oil Heater Relay
		Q5	Oil Feed Solenoid Comp
		Q6	SPARE
		Q7	Fast Unloading Coil Comp
		Q8	SPARE
		DO1	Loading Coil (V1)
		DO2	Unloading Coil (V2)
	POL985	Q1	Oil Pump Comp 1
		Q2	Cooling Tower Fan
		Q3	Condenser Pump On/Off
		Q4	Evaporator Pump On/Off
		Q5	SPARE
		Q6	SPARE
		Q7	SPARE
		Q8	SPARE
		DO1	SPARE
		DO2	SPARE

Controller T-IP Ethernet	PVR 7 INCH Communication Port
POL688	Main Board
POL985	Expansion Board